Lecture Notes

on Eigenvalues, and Long-Term population ratios

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Question: Find the long-term population structure for the population represented by the Leslie matrix

$$A = \begin{pmatrix} 1 & 4 \\ 0.5 & 0 \end{pmatrix}.$$

$|A - \lambda I| = 0,$

$\Rightarrow \begin{vmatrix} 1-\lambda & 4 \\ 0.5 & 0-\lambda \end{vmatrix} = 0$

$\Rightarrow (1-\lambda)(0-\lambda) - (0.5 \cdot 4) = 0$

$\Rightarrow 0 - \lambda + \lambda^2 - 2 = 0$

$\Rightarrow \lambda^2 - \lambda - 2 = 0$

$\Rightarrow (\lambda - 2)(\lambda + 1) = 0$

$\Rightarrow \lambda = 2, -1$

So the dominant eigenvalue is $\lambda_1 = 2$. Thus the corresponding eigenvector is given: say $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. Then,

$(A - \lambda I)\vec{x} = \vec{0}$

$\Rightarrow \begin{pmatrix} 1-2 & 4 \\ 0.5 & 0-2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$

$\Rightarrow \begin{pmatrix} -1 & 4 \\ .5 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\begin{cases} -x_1 + 4x_2 = 0 \\ \frac{1}{2}x_1 - 2x_2 = 0 \end{cases} \Rightarrow \begin{matrix} x_1 = 4x_2 \\ x_1 = 4x_2 \end{matrix} \Rightarrow x_1 = 4x_2$

Let x_1 be 1,

$x_2 = \frac{1}{4}$

$\therefore \vec{x} = \begin{pmatrix} 1 \\ 1/4 \end{pmatrix} \Rightarrow 4\vec{x} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

Thus the corresponding normalized eigenvector is given by

$$\vec{v} = \begin{pmatrix} 4/5 \\ 1/5 \end{pmatrix} \quad \left\{ \text{or } \begin{pmatrix} 1/(1+1/4) \\ \frac{1}{4}/(1+1/4) \end{pmatrix} = \begin{pmatrix} 1/\frac{5}{4} \\ \frac{1}{4}/\frac{5}{4} \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ \frac{1}{5} \end{pmatrix} \right.$$

Note anyway it will give same answer.

In a long run the population distribution are given by

$\frac{4}{5} \times 100\% = 80\%$ and $\frac{1}{5} \times 100\% = 20\%$.